

Time: 30 minutes

Max marks: 10

Instructions:

- Do not plagiarize. Do not assist your classmates in plagiarism.
- Define all your variables properly, and show your full solution to get full credit.
- Attempt all questions that you can.
- In the unlikely case a question is not clear, discuss it with an invigilating TA. Please ensure that you clearly include any assumptions you make, even after clarification from the invigilator.

1. Let $\mathbf{A}$ be an $m \times n$ matrix of rank r , and its SVD given by $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$. Answer the following questions:
- ($1/2$ point) Explain how is the column space of $\mathbf{A}$ related to that of $\mathbf{A}\mathbf{A}^\top$.
 - ($1/2$ point) Explain how are the column vectors of $\mathbf{U}$ related to the column space of $\mathbf{A}$.
 - (2 points) Identify a set of orthogonal bases for each of the four fundamental subspaces of $\mathbf{A}$ using the column vectors of $\mathbf{U}$ and $\mathbf{V}$.

Total for Question 1: 3

Solution:

- (a) ($1/2$ point) **Key idea:** $Col(\mathbf{A}\mathbf{A}^\top) = Col(\mathbf{A})$.

Reason: For any vector $\mathbf{z}$, $\mathbf{A}\mathbf{A}^\top \mathbf{z} = \mathbf{A}(\mathbf{A}^\top \mathbf{z}) \in Col(\mathbf{A})$, hence $Col(\mathbf{A}\mathbf{A}^\top) \subseteq Col(\mathbf{A})$. Also, using SVD,

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \Rightarrow \mathbf{A}\mathbf{A}^\top = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \mathbf{V}\mathbf{\Sigma}^\top \mathbf{U}^\top = \mathbf{U}(\mathbf{\Sigma}\mathbf{\Sigma}^\top)\mathbf{U}^\top.$$

Since $rank(\mathbf{A}) = r$, $\mathbf{\Sigma}\mathbf{\Sigma}^\top$ has exactly r positive diagonal entries, so $Col(\mathbf{A}\mathbf{A}^\top)$ is spanned by the first r columns of $\mathbf{U}$, which is also $Col(\mathbf{A})$. Therefore $Col(\mathbf{A}\mathbf{A}^\top) = Col(\mathbf{A})$.

Rubric:

0.25: States $Col(\mathbf{A}\mathbf{A}^\top) \subseteq Col(\mathbf{A})$ with correct reasoning ($\mathbf{A}(\cdot)$ form).

0.25: Concludes equality with a correct argument (via SVD / same left singular subspace / same span of first r columns of $\mathbf{U}$).

0: Incorrect relationship or no justification.

- (b) ($1/2$ point) For $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ with $rank(\mathbf{A}) = r$, the first r columns $\mathbf{u}_1, \dots, \mathbf{u}_r$ of $\mathbf{U}$ (those corresponding to nonzero singular values) form an orthonormal basis for $Col(\mathbf{A})$. The remaining columns $\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$ span $Null(\mathbf{A}^\top)$.

Rubric:

0.25: Correctly identifies that $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$ span $Col(\mathbf{A})$ (must specify first r /nonzero singular values).

0.25: Mentions orthonormal basis nature (or correctly notes remaining columns span $Null(\mathbf{A}^\top)$).

0: Vague/incorrect statement.

- (c) (2 points) The four fundamental subspaces and orthogonal bases from the SVD:

- **Column space** $Col(\mathbf{A}) \subseteq \mathbb{R}^m$: orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$.
- **Left null space** $Null(\mathbf{A}^\top) \subseteq \mathbb{R}^m$: orthonormal basis $\{\mathbf{u}_{r+1}, \dots, \mathbf{u}_m\}$.
- **Row space** $Row(\mathbf{A}) = Col(\mathbf{A}^\top) \subseteq \mathbb{R}^n$: orthonormal basis $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$.
- **Null space** $Null(\mathbf{A}) \subseteq \mathbb{R}^n$: orthonormal basis $\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}$.

Rubric:0.5: Correct basis for $\text{Col}(\mathbf{A})$ as $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$.0.5: Correct basis for $\text{Null}(\mathbf{A}^\top)$ as $\{\mathbf{u}_{r+1}, \dots, \mathbf{u}_m\}$.0.5: Correct basis for $\text{Row}(\mathbf{A})$ as $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$.0.5: Correct basis for $\text{Null}(\mathbf{A})$ as $\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}$.**Common partial credit:**

Deduct 0.25 for each subspace that is mislabeled but the correct vector set is given (e.g., row/column swapped).

0: If no correct mapping to any of the four subspaces.

2. The linear regression model for a d -dimensional variable $\mathbf{x}$ and its corresponding target variable y is given by

$$y = \mathbf{x}^\top \boldsymbol{\theta} + \epsilon$$

where, $\epsilon \sim \mathcal{N}(0, \sigma^2)$, i.e., ϵ is random error drawn from a zero-mean Gaussian distribution. Answer the following questions:

- ($1/2$ point) For N training samples $(\mathbf{x}^{(i)}, y^{(i)})$, $i = 1, \dots, N$, write the least-squares loss function for linear regression.
- ($1/2$ point) Rewrite the loss in matrix form using appropriate norm functions.
- (2 points) Derive the solution for the optimal set of parameters $\boldsymbol{\theta}^*$ that solves the least-squares problem.

Total for Question 2: 3

Solution:

- (a) ($1/2$ point) The least-squares loss is

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^N \left(y^{(i)} - (\mathbf{x}^{(i)})^\top \boldsymbol{\theta} \right)^2.$$

Rubric:

0.5: Correct sum of squared residuals with correct indexing and squared term.

0.25: If summation bounds are missing but the expression is clearly LS over all samples.

0: If absolute loss / wrong model / missing square.

- (b) ($1/2$ point) Define $\mathbf{X} \in \mathbb{R}^{N \times d}$ with i -th row $(\mathbf{x}^{(i)})^\top$ and $\mathbf{y} \in \mathbb{R}^N$ with entries $y^{(i)}$. Then

$$\mathcal{L}(\boldsymbol{\theta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 \quad (\text{equivalently } (\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\theta})).$$

Rubric:0.25: Correct matrix setup for $\mathbf{X}$ and $\mathbf{y}$ (rows as samples, correct dimensions).0.25: Correct matrix-form loss $\|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2$ (or equivalent quadratic form).

0: Wrong norm / wrong multiplication order.

- (c) (2 points) Start with

$$\mathcal{L}(\boldsymbol{\theta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\theta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\theta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\theta}.$$

Differentiate and set gradient to zero:

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \boldsymbol{\theta} = \mathbf{0} \Rightarrow \mathbf{X}^\top \mathbf{X} \boldsymbol{\theta}^* = \mathbf{X}^\top \mathbf{y}.$$

If $\mathbf{X}^\top \mathbf{X}$ is invertible (full column rank),

$$\boldsymbol{\theta}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

If not invertible, an acceptable solution is the Moore–Penrose pseudoinverse:

$$\boldsymbol{\theta}^* = \mathbf{X}^\dagger \mathbf{y}.$$

Rubric:

0.5: Correct normal equations $\mathbf{X}^\top \mathbf{X} \boldsymbol{\theta}^* = \mathbf{X}^\top \mathbf{y}$.

0.5: Correct derivation step (gradient / differentiation shown).

0.5: Correct closed form $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ with condition stated (invertible/full rank).

0.5: Mentions pseudoinverse / non-unique solution when $\mathbf{X}^\top \mathbf{X}$ singular (acceptable alternative).

0: If final expression is wrong or no derivation.

Alternative acceptable answers:

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2, \quad \boldsymbol{\theta}^* = \mathbf{X}^\dagger \mathbf{y}.$$

3. (a) (1 point) Write the binary cross-entropy loss function, which is used for training a logistic regression problem.
- (b) (1/2 point) What are the minimum and maximum values that this loss function can take?
- (c) (1/2 point) How would you modify this loss function to train a multi-class classification variant?
- (d) (1 point) After training my classifier, my training accuracy is 98% and my validation accuracy is 65%. What can you say about my model? Is it properly trained? If yes, explain why, else suggest a change in my training strategy.
- (e) (1 point) For a different setting, my training accuracy is 65% and my validation accuracy is 98%. What can you say about this model? What would you suggest to change?

Total for Question 3: 4

Solution:

- (a) (1 point) For binary labels $y \in \{0, 1\}$ and predicted probability $\hat{y}$, the binary cross-entropy loss over N samples is

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N \left[y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right].$$

(A sum without the $\frac{1}{N}$ is also acceptable.)

Rubric:

0.5: Correct BCE per-sample form $-y \log \hat{y} - (1 - y) \log(1 - \hat{y})$.

0.5: Correct aggregation (sum/mean) with correct indices/notation.

0.25: If missing minus sign but clearly identifies “negative log-likelihood”.

0: If uses MSE or incorrect expression.

- (b) (1/2 point) **Minimum:** 0 (achieved when predictions are perfect).
Maximum: unbounded / $+\infty$ (can diverge when the model is confidently wrong since $\log(0) \rightarrow -\infty$).

Rubric:

0.25: Correct minimum value 0.

0.25: Correct maximum value unbounded/ $+\infty$.

0: If states bounded max or wrong values.

- (c) ($\frac{1}{2}$ point) For K classes, use softmax cross-entropy. With one-hot $\mathbf{y}^{(i)} \in \{0, 1\}^K$, $\|\mathbf{y}^{(i)}\|_1 = 1$ and $\hat{\mathbf{y}}^{(i)} = \text{softmax}(\mathbf{z}^{(i)})$:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_k^{(i)} \log \hat{y}_k^{(i)}.$$

(Equivalently, $-\frac{1}{N} \sum_i \log \hat{y}_{\text{true class}}^{(i)}$.)

Rubric:

0.25: Mentions softmax + categorical cross-entropy / multi-class NLL.

0.25: Correct formula with summation over classes (or $-\log p_{\text{true}}$).

0: If suggests BCE independently per class (unless explicitly framed as multi-label).

- (d) (1 point) Train acc 98% and validation acc 65% indicates **overfitting** (high variance / poor generalization). The model fits training data well but does not generalize.

Fix: add regularization (L2/weight decay), dropout, early stopping, data augmentation, reduce model capacity, or collect more data.

Rubric:

0.5: Correct diagnosis: overfitting / high variance / poor generalization.

0.5: At least one correct remedy (regularization/augmentation/early stopping/reduce capacity/more data).

0: If calls it underfitting or says it is properly trained without issues.

- (e) (1 point) Train acc 65% and validation acc 98% is **unusual/suspicious**. Likely causes include **data leakage** (validation info leaking into training or preprocessing fit on full data), an **evaluation bug** (wrong split/metric), or that the validation set is much easier / distribution mismatch.

Fix: verify the split, remove duplicates across splits, ensure preprocessing/statistics are fit only on training, recompute accuracy correctly, and redo a proper (e.g., stratified) split.

Rubric:

0.5: Flags it as suspicious and gives a correct cause (leakage/bug/split issue/easier validation).

0.5: Provides at least one concrete corrective action (check split, leakage, duplicates, preprocessing, metric).

0: If claims it is a great model without caveats.